INTEGRAL REPRESENTATIONS OF THE SUBELLIPTIC HEAT KERNEL OF THE OCTONIONIC ANTI-DE SITTER FIBRATION

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Allowed standard prerequisites (editorial).

- Original author-cited octonionic Hopf/complex/quaternionic and Jacobi/Bergman wave-kernel background. Full original bibliography and related-work distinctions retained. Complete source-local geometry and integral proofs retained.

1. Introduction and results

In this note we study the sub-Laplacian and the corresponding sub-Riemannian heat kernel of the octonionic anti-de Sitter fibration

$$\mathbb{S}^7 \hookrightarrow \mathrm{AdS}^{15}(\mathbb{O}) \to \mathbb{O}H^1.$$

This paper follows the previous works [2, 3, 10] which respectively concerned:

(1) *The complex anti-de Sitter fibrations*:

$$\mathbb{S}^1 \hookrightarrow \mathrm{AdS}^{2n+1}(\mathbb{C}) \to \mathbb{C}H^n.$$

(2) *The quaternionic anti-de Sitter fibrations*:

$$\mathbb{S}^3 \hookrightarrow \mathrm{AdS}^{4n+3}(\mathbb{H}) \to \mathbb{H}H^n.$$

The 15-dimensional anti-de Sitter fibration is the last model space that remained to be studied of a sub-Riemannian manifold arising from a H-type semi-Riemannian submersion over a rank-one symmetric space, see the Table 3 in [4].

Similarly to the complex and quaternionic case, the sub-Laplacian is defined as the lift on $\mathrm{AdS}^{15}(\mathbb{O})$ of the Laplace–Beltrami operator of the octonionic hyperbolic space $\mathbb{O}H^1$. However, in the complex and quaternionic case the Lie group structure of the fiber played an important role that we can not use here, since the fiber $\mathbb{S}^7$ is not a group. Instead, we make use of some algebraic properties of $\mathbb{S}^7$ that were already pointed out and used by the authors in [1] for the study of the octonionic Hopf fibration:

$$\mathbb{S}^7 \hookrightarrow \mathbb{S}^{15} \to \mathbb{O}P^1.$$

Let us briefly describe our main results. Due to the cylindrical symmetries of the fibration, the heat kernel of the sub-Laplacian only depends on two variables: the variable r which is the Riemannian distance on $\mathbb{O}H^1$ (the starting point is specified with inhomogeneous coordinate in Section 3) and the variable η which is the Riemannian distance starting at a pole on the fiber $\mathbb{S}^7$. We prove in Proposition 3.1 that in these coordinates, the radial part of the sub-Laplacian $\tilde{L}$ writes

$$\tilde{L} = \frac{\partial^2}{\partial r^2} + (7\coth r + 7\tanh r)\frac{\partial}{\partial r} + \tanh^2 r \left(\frac{\partial^2}{\partial \eta^2} + 6\cot\eta \frac{\partial}{\partial \eta}\right).$$

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As a consequence of this expression for the sub-Laplacian, we are able to derive two equivalent formulas for the heat kernel. The first formula, see Proposition 4.1, reads as follows: for $r \geq 0$, $\eta \in [0, \pi)$, $t > 0$

$$p_t(r, \eta) = \int_0^\infty s_t(\eta, \mathrm{i}u) q_{t,15}(\cosh r \cosh u) \sinh^6 u \, \mathrm{d}u,$$

where s_t is the heat kernel of the Jacobi operator

$$\tilde{\triangle}_{\mathbb{S}^7} = \frac{\partial^2}{\partial \eta^2} + 6 \cot \eta \frac{\partial}{\partial \eta}$$

with respect to the measure $\sin^6 \eta \, \mathrm{d}\eta$, and where $q_{t,15}$ is the Riemannian heat kernel on the 15-dimensional real hyperbolic space $\mathbb{H}^{15}$ given in (4.1). The second formula, see Proposition 4.2, writes as follows:

$$p_t(r, \eta) = \int_0^\pi \int_0^\infty G_t(\eta, \varphi, u) q_{t,9}(\cosh r \cosh u) \sin^5 \varphi \, \mathrm{d}u \, \mathrm{d}\varphi,$$

where $q_{t,9}$ is Riemannian heat kernel on the 9-dimensional hyperbolic space $\mathbb{H}^9$ and $G_t(\eta, \varphi, u)$ is given in (4.3).

Similarly to [2, 3, 10], it might be expected that explicit integral representations of the heat kernel might be used to study small-time asymptotics, inside and outside of the cut-locus. Integral representations of heat kernels can also be used to obtain sharp heat kernel estimates, see [7]. Those applications of the heat kernel representations we obtain will possibly be addressed in a future research project.

2. The octonionic anti-de Sitter fibration

Let

$$\mathbb{O} = \left\{ x = \sum_{j=0}^7 x_j e_j, \, x_j \in \mathbb{R} \right\},$$

be the division algebra of octonions (see [9] for explicit representations of this algebra). We recall that the multiplication rules are given by

$$\begin{aligned} e_i e_j &= e_j \qquad \text{if} \quad i = 0, \\ e_i e_j &= e_i \qquad \text{if} \quad j = 0, \\ e_i e_j &= -\delta_{ij} e_0 + \epsilon_{ijk} e_k \qquad \text{otherwise}, \end{aligned}$$

where δ_{ij} is the Kronecker delta and ϵ_{ijk} is the completely antisymmetric tensor with value 1 when $ijk = 123, 145, 176, 246, 257, 347, 365$ (also see [1]). The octonionic norm is defined for $x \in \mathbb{O}$ by

$$||x||^2 = \sum_{j=0}^7 x_j^2.$$

The octonionic anti-de Sitter space $\mathrm{AdS}^{15}(\mathbb{O})$ is the quadric defined as the pseudo-hyperbolic space by:

$$\mathrm{AdS}^{15}(\mathbb{O}) = \left\{ (x, y) \in \mathbb{O}^2, \, ||(x, y)||_{\mathbb{O}}^2 = -1 \right\},$$

where

$$||(x, y)||_{\mathbb{O}}^2 := ||x||^2 - ||y||^2.$$

In real coordinates we have $x = \sum_{j=0}^7 x_j e_j$, $y = \sum_{j=0}^7 y_j e_j$, and the pseudo-norm can be written as

$$x_0^2 + \cdots + x_7^2 - y_0^2 - \cdots - y_7^2.$$

As such, $\mathrm{AdS}^{15}(\mathbb{O})$ is embedded in the flat 16-dimensional space $\mathbb{R}^{8,8}$ endowed with the Lorentzian real signature $(8,8)$ metric

$$\mathrm{d}s^2 = \mathrm{d}x_0^2 + \cdots + \mathrm{d}x_7^2 - \mathrm{d}y_0^2 - \cdots - \mathrm{d}y_7^2.$$

Consequently, $\mathrm{AdS}^{15}(\mathbb{O})$ is naturally endowed with a pseudo-Riemannian structure of signature $(8,7)$.

Let $\mathbb{O}H^1$ denote the octonionic hyperbolic space. The map $\pi\colon \mathrm{AdS}^{15}(\mathbb{O}) \to \mathbb{O}H^1$, given by $(x,y) \mapsto [x:y] = y^{-1}x$ is a pseudo-Riemannian submersion with totally geodesic fibers isometric to the seven-dimensional sphere $\mathbb{S}^7$. Notice that, as a topological manifold, $\mathbb{O}H^1$ can therefore be identified with the unit open ball in $\mathbb{O}$. The pseudo-Riemannian submersion π yields the octonionic anti-de Sitter fibration

$$\mathbb{S}^7 \hookrightarrow \mathrm{AdS}^{15}(\mathbb{O}) \to \mathbb{O}H^1.$$

For further information on semi-Riemannian submersions over rank-one symmetric spaces, we refer to [6].

3. Cylindrical coordinates and radial part of the sub-Laplacian

The sub-Laplacian L on $\mathrm{AdS}^{15}(\mathbb{O})$ we are interested in is the horizontal Laplacian of the Riemannian submersion $\pi\colon \mathrm{AdS}^{15}(\mathbb{O}) \to \mathbb{O}H^1$, i.e., the horizontal lift of the Laplace–Beltrami operator of $\mathbb{O}H^1$. It can be written as

$$L = \Box_{\mathrm{AdS}^{15}(\mathbb{O})} + \triangle_{\mathcal{V}}, \tag{3.1}$$

where $\Box_{\mathrm{AdS}^{15}(\mathbb{O})}$ is the d'Alembertian, i.e., the Laplace–Beltrami operator of the pseudo-Riemannian metric and $\triangle_{\mathcal{V}}$ is the vertical Laplacian. Since the fibers of π are totally geodesic and isometric to $\mathbb{S}^7 \subset \mathrm{AdS}^{15}(\mathbb{O})$, we note that $\Box_{\mathrm{AdS}^{15}(\mathbb{O})}$ and $\triangle_{\mathcal{V}}$ are commuting operators, and we can identify

$$\triangle_{\mathcal{V}} = \triangle_{\mathbb{S}^7}. \tag{3.2}$$

The sub-Laplacian L is associated with a canonical sub-Riemannian structure on $\mathrm{AdS}^{15}(\mathbb{O})$ which is of H-type, see [4].

To study L, we introduce a set of coordinates that reflect the cylindrical symmetries of the octonionic unit sphere which provides an explicit local trivialization of the octonionic anti-de Sitter fibration. Consider the coordinates $w \in \mathbb{O}H^1$, where w is the inhomogeneous coordinate on $\mathbb{O}H^1$ given by $w = y^{-1}x$, with $x, y \in \mathrm{AdS}^{15}(\mathbb{O})$. Consider the north pole $p \in \mathbb{S}^7$ and take $Y_1, \dots, Y_7$ to be an orthonormal frame of $T_p\mathbb{S}^7$. Let us denote $\exp_p$ the Riemannian exponential map at p on $\mathbb{S}^7$. Then the cylindrical coordinates we work with are given by

$$(w, \theta_1, \dots, \theta_7) \mapsto \left(\frac{\exp_p\left(\sum_{i=1}^7 \theta_i Y_i\right) w}{\sqrt{1-\rho^2}}, \frac{\exp_p\left(\sum_{i=1}^7 \theta_i Y_i\right)}{\sqrt{1-\rho^2}} \right) \in \mathrm{AdS}^{15}(\mathbb{O}),$$

where $\rho = \|w\|$ and $\|\theta\| = \sqrt{\theta_1^2 + \cdots + \theta_7^2} < \pi$.

A function f on $\mathrm{AdS}^{15}(\mathbb{O})$ is called radial cylindrical if it only depends on the two coordinates $(\rho, \eta) \in [0,1) \times [0,\pi]$ where $\eta = \sqrt{\sum_{i=1}^7 \theta_i^2}$. More precisely f is radial cylindrical if there exists a function g so that

$$f\left(\frac{\exp_p\left(\sum_{i=1}^7 \theta_i Y_i\right) w}{\sqrt{1-\rho^2}}, \frac{\exp_p\left(\sum_{i=1}^7 \theta_i Y_i\right)}{\sqrt{1-\rho^2}} \right) = g(\rho, \eta).$$

We denote by $\mathcal{D}$ the space of smooth and compactly supported functions on $[0,1) \times [0,\pi)$. Then the radial part of L is defined as the operator $\widetilde{L}$ such that for any $f \in \mathcal{D}$, we have

$$L(f \circ \psi) = \left(\widetilde{L} f\right) \circ \psi. \tag{3.3}$$

We now compute $\widetilde{L}$ in cylindrical coordinates.

Proposition 3.1. *The radial part of the sub-Laplacian on* $\mathrm{AdS}^{15}(\mathbb{O})$ *is given in the coordinates* (r,η) *by the operator*

$$\widetilde{L} = \frac{\partial^2}{\partial r^2} + (7\coth r + 7\tanh r)\frac{\partial}{\partial r} + \tanh^2 r\left(\frac{\partial^2}{\partial \eta^2} + 6\cot\eta\frac{\partial}{\partial \eta}\right),$$

where $r = \tanh^{-1}\rho$ *is the Riemannian distance on* $\mathbb{O}H^1$ *from the origin.*

Proof. Note that the radial part of the Laplace–Beltrami operator on the octonionic hyperbolic space $\mathbb{O}H^1$ is

$$\widetilde{\triangle}_{\mathbb{O}H^1} = \frac{\partial^2}{\partial r^2} + (7\coth r + 7\tanh r)\frac{\partial}{\partial r},$$

and the radial part of the Laplace–Beltrami operator on $\mathbb{S}^7$ is

$$\widetilde{\triangle}_{\mathbb{S}^7} = \frac{\partial^2}{\partial \eta^2} + 6\cot\eta\frac{\partial}{\partial \eta}. \tag{3.4}$$

Since the octonionic anti-de Sitter fibration defines a totally geodesic submersion with base space $\mathbb{O}H^1$ and fiber $\mathbb{S}^7$, the semi-Riemannian metric on $\mathrm{AdS}^{15}(\mathbb{O})$ is locally given by a warped product between the Riemannian metric of $\mathbb{O}H^1$ and the Riemannian metric on $\mathbb{S}^7$. Hence the radial part of the d'Alembertian becomes

$$\widetilde{\Box}_{\mathrm{AdS}^{15}(\mathbb{O})} = \frac{\partial^2}{\partial r^2} + (7\coth r + 7\tanh r)\frac{\partial}{\partial r} + g(r)\left(\frac{\partial^2}{\partial \eta^2} + 6\cot\eta\frac{\partial}{\partial \eta}\right), \tag{3.5}$$

for some smooth function g to be computed.

On the other hand, from the isometric embedding $\mathrm{AdS}^{15}(\mathbb{O}) \subset \mathbb{O}\times\mathbb{O}$, the d'Alembertian on $\mathrm{AdS}^{15}(\mathbb{O})$ is a restriction of the d'Alembertian on $\mathbb{O}\times\mathbb{O} \simeq \mathbb{R}^{8,8}$ in the sense that for a smooth $f\colon \mathrm{AdS}^{15}(\mathbb{O}) \to \mathbb{R}$

$$\Box_{\mathrm{AdS}^{15}(\mathbb{O})} f = \Box_{\mathbb{O}\times\mathbb{O}} f^*_{/\mathrm{AdS}^{15}(\mathbb{O})},$$

where $\Box_{\mathbb{O}\times\mathbb{O}} = \sum_{i=0}^{7}\left(\frac{\partial^2}{\partial x_i^2} - \frac{\partial^2}{\partial y_i^2}\right)$ and for $x,y \in \mathbb{O}$ such that $\|y\|^2 - \|x\|^2 > 0$, $f^*(x,y) = f\big(\frac{x}{\sqrt{\|y\|^2-\|x\|^2}}, \frac{y}{\sqrt{\|y\|^2-\|x\|^2}}\big)$. For the specific choice of the function $f(x,y) = y_1$, one easily computes that $\Box_{\mathbb{O}\times\mathbb{O}} f^*_{/\mathrm{AdS}^{15}(\mathbb{O})}(x,y) = 15y_1$, thus

$$\Box_{\mathrm{AdS}^{15}(\mathbb{O})} f(x,y) = 15y_1.$$

For the point with coordinates

$$\left(\frac{\exp_p\left(\sum_{i=1}^{7}\theta_i Y_i\right)w}{\sqrt{1-\rho^2}}, \frac{\exp_p\left(\sum_{i=1}^{7}\theta_i Y_i\right)}{\sqrt{1-\rho^2}}\right) \in \mathrm{AdS}^{15}(\mathbb{O})$$

one has

$$y_1 = \frac{\cos\eta}{\sqrt{1-\rho^2}} = \cosh r\cos\eta.$$

We therefore deduce that

$$\widetilde{\Box}_{\mathrm{AdS}^{15}(\mathbb{O})}(\cosh r\cos\eta) = 15\cosh r\cos\eta.$$

Using the formula (3.5), after a straightforward computation, this yields $g(r) = -\frac{1}{\cosh^2 r}$ and therefore

$$\begin{aligned}\widetilde{\Box}_{\mathrm{AdS}^{15}(\mathbb{O})} &= \frac{\partial^2}{\partial r^2} + (7\coth r + 7\tanh r)\frac{\partial}{\partial r} - \frac{1}{\cosh^2 r}\left(\frac{\partial^2}{\partial \eta^2} + 6\cot\eta\frac{\partial}{\partial \eta}\right)\\ &= \widetilde{\triangle}_{\mathbb{O}H^1} - \frac{1}{\cosh^2 r}\widetilde{\triangle}_{\mathbb{S}^7}.\end{aligned}$$

Finally, to conclude, one notes that the sub-Laplacian L is given by the difference between the Laplace–Beltrami operator of $\mathrm{AdS}^{15}(\mathbb{O})$ and the vertical Laplacian. Therefore by (3.1) and (3.2),

□

$$\widetilde{L} = \tilde{\Box}_{\mathrm{AdS}^{15}(\mathbb{O})} + \tilde{\triangle}_{\mathbb{S}^7} = \frac{\partial^2}{\partial r^2} + (7\coth r + 7\tanh r)\frac{\partial}{\partial r} + \tanh^2 r\left(\frac{\partial^2}{\partial \eta^2} + 6\cot\eta\frac{\partial}{\partial \eta}\right).$$

Remark 3.2. As a consequence of the previous result, we can check that the Riemannian measure of $\mathrm{AdS}^{15}(\mathbb{O})$ in the coordinates (r,η), which is the symmetric and invariant measure for $\tilde{L}$ is given by

$$\mathrm{d}\overline{\mu} = \frac{\pi^7}{90}\sinh^7 r\cosh^7 r\sin^6\eta\,\mathrm{d}r\,\mathrm{d}\eta. \tag{3.6}$$

(See also Remark 2 in [1], which corresponds to the case of the octonionic Hopf fibration.)

4. Integral representations of the subelliptic heat kernel

In this section, we give two integral representations of the subelliptic heat kernel associated with $\tilde{L}$. We denote by $p_t(r,\eta)$ the heat kernel of $\tilde{L}$ issued from the point $r=\eta=0$ with respect to the measure (3.6). We remark that studying the subelliptic heat kernel associated with $\tilde{L}$ is enough to study the heat kernel of L, because due to (3.3) the heat kernel $h_t(w,\theta)$ of L issued from the point with cylindric coordinates $w=0$, $\theta=0$ is then given by

$$h_t(w,\theta) = p_t\big(\tanh^{-1}\|w\|, \|\theta\|\big).$$

4.1. **First integral representation.** We denote by s_t the heat kernel of the operator

$$\tilde{\triangle}_{\mathbb{S}^7} = \frac{\partial^2}{\partial\eta^2} + 6\cot\eta\frac{\partial}{\partial\eta}$$

with respect to the reference measure $\sin^6\eta\,\mathrm{d}\eta$. The operator $\tilde{\triangle}_{\mathbb{S}^7}$ belongs to the family of Jacobi diffusion operators which have been extensively studied in the literature, see for instance the appendix in [5] and the references therein. In particular, the spectrum of $\tilde{\triangle}_{\mathbb{S}^7}$ is given by

$$\mathbf{Sp}\big(-\tilde{\triangle}_{\mathbb{S}^7}\big) = \{m(m+6),\, m\in\mathbb{N}\},$$

and the eigenfunction corresponding to the eigenvalue $m(m+6)$ is $P_m^{5/2,5/2}(\cos\eta)$ where $P_m^{5/2,5/2}$ is the Jacobi polynomial

$$P_m^{5/2,5/2}(x) = \frac{(-1)^m}{2^m m!\big(1-x^2\big)^{5/2}}\frac{\mathrm{d}^m}{\mathrm{d}x^m}\big(1-x^2\big)^{5/2+m}.$$

As a consequence, one has the following spectral decomposition for the heat kernel:

$$s_t(\eta,u) = \frac{1}{\pi}\sum_{m=0}^{+\infty}\frac{2^{4m+7}m!(m+5)![(m+3)!]^2}{(2m+6)!(2m+5)!}\mathrm{e}^{-m(m+6)t}P_m^{5/2,5/2}(\cos\eta)P_m^{5/2,5/2}(\cos u).$$

Proposition 4.1. *For $r\geq 0$, $\eta\in[0,\pi]$, and $t>0$ we have*

$$p_t(r,\eta) = \int_0^\infty s_t(\eta,\mathrm{i}u)q_{t,15}(\cosh r\cosh u)\sinh^6 u\,\mathrm{d}u,$$

where

$$q_{t,15}(\cosh s) := \frac{\mathrm{e}^{-49t}}{(2\pi)^7\sqrt{4\pi t}}\left(-\frac{1}{\sinh s}\frac{\mathrm{d}}{\mathrm{d}s}\right)^7\mathrm{e}^{-s^2/4t} \tag{4.1}$$

is the Riemannian heat kernel on the 15-dimensional real hyperbolic space $\mathbb{H}^{15}$.

Proof. Since $\pi\colon \mathrm{AdS}^{15}(\mathbb{O}) \to \mathbb{O}H^1$ is a (semi-Riemannian) totally geodesic submersion, the operators $\widetilde{\square}_{\mathrm{AdS}^{15}(\mathbb{O})}$ and $\widetilde{\triangle}_{\mathbb{S}^7}$ commute. Thus

$$\mathrm{e}^{t\widetilde{L}} = \mathrm{e}^{t(\widetilde{\square}_{\mathrm{AdS}^{15}(\mathbb{O})}+\widetilde{\triangle}_{\mathbb{S}^7})} = \mathrm{e}^{t\widetilde{\triangle}_{\mathbb{S}^7}}\mathrm{e}^{t\widetilde{\square}_{\mathrm{AdS}^{15}(\mathbb{O})}}.$$

We deduce that the heat kernel of $\tilde{L}$ can be written as

$$p_t(r,\eta) = \int_0^\pi s_t(\eta,u) p_t^{\widetilde{\square}_{\mathrm{AdS}^{15}(\mathbb{O})}}(r,u)\sin^6 u\,\mathrm{d}u, \tag{4.2}$$

where s_t is the heat kernel of (3.4) with respect to the measure $\sin^6\eta\,\mathrm{d}\eta$, $\eta\in[0,\pi)$, and $p_t^{\widetilde{\square}_{\mathrm{AdS}^{15}(\mathbb{O})}}(r,u)$ the heat kernel at $(0,0)$ of $\widetilde{\square}_{\mathrm{AdS}^{15}(\mathbb{O})}$ with respect to the measure in (3.6), i.e.,

$$\mathrm{d}\mu(r,u) = \frac{\pi^7}{90}\sinh^7 r\cosh^7 r\sin^6 u\,\mathrm{d}r\,\mathrm{d}u, \qquad r\in[0,\infty), \qquad u\in[0,\pi].$$

In order to write (4.2) more precisely, let us consider the analytic change of variables $\tau\colon (r,\eta)\to(r,\mathrm{i}\eta)$ that will be applied on functions of the type $f(r,\eta)=h(r)\mathrm{e}^{-\mathrm{i}\lambda\eta}$, with h smooth and compactly supported on $[0,\infty)$ and $\lambda>0$. Then as we saw in the proof of Proposition 3.1 one can see that

$$\widetilde{\square}_{\mathrm{AdS}^{15}(\mathbb{O})}(f\circ\tau) = \big(\widetilde{\triangle}_{\mathbb{H}^{15}} f\big)\circ\tau,$$

where

$$\widetilde{\triangle}_{\mathbb{H}^{15}} = \widetilde{\triangle}_{\mathbb{O}H^1} + \frac{1}{\cosh^2 r}\widetilde{\triangle}_P, \qquad \widetilde{\triangle}_P = \frac{\partial^2}{\partial\eta^2} + 6\coth\eta\frac{\partial}{\partial\eta}.$$

Then, one deduces

$$\mathrm{e}^{t\widetilde{L}}(f\circ\tau) = \mathrm{e}^{t\widetilde{\triangle}_{\mathbb{S}^7}}\mathrm{e}^{t\widetilde{\square}_{\mathrm{AdS}^{15}(\mathbb{O})}}(f\circ\tau) = \mathrm{e}^{t\widetilde{\triangle}_{\mathbb{S}^7}}\big(\big(\mathrm{e}^{t\widetilde{\triangle}_{\mathbb{H}^{15}}} f\big)\circ\tau\big) = \big(\mathrm{e}^{-t\widetilde{\triangle}_P}\mathrm{e}^{t\widetilde{\triangle}_{\mathbb{H}^{15}}} f\big)\circ\tau.$$

Now, since for every $f(r,\eta)=h(r)\mathrm{e}^{-\mathrm{i}\lambda\eta}$,

$$\big(\mathrm{e}^{t\widetilde{\square}_{\mathrm{AdS}^{15}(\mathbb{O})}} f\big)(0,0) = \big(\mathrm{e}^{t\widetilde{\triangle}_{\mathbb{H}^{15}}}\big)\big(f\circ\tau^{-1}\big)(0,0),$$

one deduces that for a function h depending only on u,

$$\int_0^\pi h(u) p_t^{\widetilde{\square}_{\mathrm{AdS}^{15}(\mathbb{O})}}(r,u)\sin^6 u\,\mathrm{d}u = \int_0^\infty h(-\mathrm{i}u) q_{t,15}(\cosh r\cosh u)\sinh^6 u\,\mathrm{d}u.$$

Therefore, coming back to (4.2), one infers that using the analytic extension of s_t one must have

$$\int_0^\pi s_t(\eta,u) p_t^{\square_{\mathrm{AdS}^{15}(\mathbb{O})}}(r,u)\sin^6 u\,\mathrm{d}u = \int_0^\infty s_t(\eta,-\mathrm{i}u) q_{t,15}(\cosh r\cosh u)\sinh^6 u\,\mathrm{d}u,$$

where $q_{t,15}$ is the Riemannian heat kernel on the real hyperbolic space $\mathbb{H}^{15}$ given in (4.1). □

4.2. **Second integral representation.**

Proposition 4.2. *For $r\geq 0$, $\eta\in[0,\pi]$, and $t>0$ we have*

$$p_t(r,\eta) = \int_0^\pi\int_0^\infty G_t(\eta,\varphi,u) q_{t,9}(\cosh r\cosh u)\sin^5\varphi\,\mathrm{d}u\,\mathrm{d}\varphi.$$

where $q_{t,9}$ is the 9-dimensional Riemannian heat kernel on the hyperbolic space $\mathbb{H}^9$:

$$q_{t,9}(\cosh s) := \frac{\mathrm{e}^{-16t}}{(2\pi)^4\sqrt{4\pi t}}\left(\frac{1}{\sinh s}\frac{\mathrm{d}}{\mathrm{d}s}\right)^4 \mathrm{e}^{-s^2/4t},$$

and

$$G_t(\eta,\varphi,u) = \frac{15}{8}\sum_{m\geq 0}\mathrm{e}^{-(m(m+6)+33)t}(\cos\eta+\mathrm{i}\sin\eta\cos\varphi)^m\cosh((m+3)u). \tag{4.3}$$

Proof. The strategy of the following method appeals to some results proved in [8]. Firstly, we decompose the subelliptic heat kernel in the η variable with respect to the basis of normalized eigenfunctions of $\tilde{\triangle}_{S^7} = \frac{\partial^2}{\partial\eta^2} + 6\cot\eta\frac{\partial}{\partial\eta}$. Accordingly,

$$p_t(r,\eta) = \sum_{m\geq 0} f_m(t,r)h_m(\eta),$$

where for each m, h_m is given by

$$h_m(\eta) = \frac{15}{16}\int_0^\pi (\cos\eta + \mathrm{i}\sin\eta\cos\varphi)^m \sin^5\varphi\,\mathrm{d}\varphi$$

and $f_m(t,\cdot)$ solves the following heat equation

$$\begin{aligned}\frac{\partial}{\partial t}f_m(t,r) &= \left(\frac{\partial^2}{\partial r^2} + (7\coth r + 7\tanh r)\frac{\partial}{\partial r} - m(m+6)\tanh^2 r\right) f_m(t,r)\\ &= \left(\frac{\partial^2}{\partial r^2} + (7\coth r + 7\tanh r)\frac{\partial}{\partial r} + \frac{m(m+6)}{\cosh^2 r} - m(m+6)\right) f_m(t,r).\end{aligned}$$

We consider then the operator

$$L_m := \frac{\partial^2}{\partial r^2} + (7\coth r + 7\tanh r)\frac{\partial}{\partial r} + \frac{m(m+6)}{\cosh^2 r} + 49,$$

which was studied in [8, p. 229]. From [8, Theorem 2], with $\alpha = 3 + \frac{m}{2}, \beta = -\frac{m}{2}$, we deduce that the solution to the wave Cauchy problem associated with the subelliptic Laplacian is given $f \in C_0^\infty\left(\mathbb{O}H^1\right)$ by

$$\cos\left(s\sqrt{-L_m}\right)(f)(w) = \frac{-\sinh s}{(2\pi)^4}\left(\frac{1}{\sinh s}\frac{\mathrm{d}}{\mathrm{d}s}\right)^4 \int_{\mathbb{O}H^1} K_m(s,w,y)f(y)\frac{\mathrm{d}y}{\left(1-||y||^2\right)^8},$$

where

$$\begin{aligned}K_m(s,w,y) = \frac{(1-\overline{\langle w,y\rangle})^{3+m/2}}{(1-\langle w,y\rangle)^{m/2}} &\frac{1}{\cosh^3(d(w,y))\sqrt{\cosh^2(s)-\cosh^2(d(w,y))}}\\ &\times {}_2F_1\left(m+3,-m-3,\frac{1}{2};\frac{\cosh(d(w,y))-\cosh(s)}{2\cosh(d(w,y))}\right),\end{aligned}$$

where ${}_2F_1$ is the Gauss hypergeometric function and $\mathrm{d}y$ stands for the Lebesgue measure in $\mathbb{R}^8$. Using the spectral formula

$$\mathrm{e}^{tL} = \frac{1}{\sqrt{4\pi t}}\int_{\mathbb{R}} \mathrm{e}^{-s^2/(4t)}\cos\left(s\sqrt{-L}\right)\mathrm{d}s,$$

which holds for any non positive self-adjoint operator, we deduce that the solution to the heat Cauchy problem associated with L_m:

$$\begin{aligned}\mathrm{e}^{tL_m}(f)(w) = \frac{\mathrm{e}^{-m(m+6)t-7^2t}}{\sqrt{4\pi t}(2\pi)^4}&\int_{\mathbb{R}}\mathrm{d}s(-\sinh s)\mathrm{e}^{-s^2/(4t)}\\ &\times\left(\frac{1}{\sinh s}\frac{\mathrm{d}}{\mathrm{d}s}\right)^4\int_{\mathbb{O}H^1} K_m(s,w,y)f(y)\frac{\mathrm{d}y}{\left(1-||y||^2\right)^8}.\end{aligned}$$

Performing integration by parts 4-times,

$$\begin{aligned}&\int_{\mathbb{R}} \mathrm{d}s(-\sinh s)\left(\frac{1}{\sinh s}\frac{\mathrm{d}}{\mathrm{d}s}\right)^4 \mathrm{e}^{-s^2/(4t)} \int_{\mathbb{O}H^1} K_m(s,w,y)f(y)\frac{\mathrm{d}y}{\left(1-||y||^2\right)^8}\\ &\quad= \int_{\mathbb{O}H^1} f(y)\frac{\mathrm{d}y}{(1-||y||^2)^8}\int_{\mathbb{R}} \mathrm{d}s(-\sinh s)K_m(s,w,y)\left(\frac{1}{\sinh s}\frac{\mathrm{d}}{\mathrm{d}s}\right)^4 \mathrm{e}^{-s^2/4t}\\ &= 2\int_{\mathbb{O}H^1} f(y)\frac{\mathrm{d}y}{\left(1-||y||^2\right)^8}\int_{d(w,y)}^{\infty} \mathrm{d}(\cosh(s))K_m(s,w,y)\left(\frac{1}{\sinh s}\frac{\mathrm{d}}{\mathrm{d}s}\right)^4 \mathrm{e}^{-s^2/4t}.\end{aligned}$$

Thus we get

$$\mathrm{e}^{tL_m}(f)(0) = 2\mathrm{e}^{-(m(m+6)+33)t}\int_{\mathbb{O}H^1} f(y)\frac{\mathrm{d}y}{\left(1-||y||^2\right)^8}\int_{d(0,y)}^{\infty} \mathrm{d}(\cosh s)K_m(s,0,y)q_{t,9}(\cosh s).$$

As a result, the subelliptic heat kernel of L_m reads

$$\begin{aligned}&\frac{\mathrm{d}y}{\left(1-||y||^2\right)^8}\int_{d(0,y)}^{\infty} \mathrm{d}(\cosh s)K_m(s,0,y)q_{t,9}(\cosh s)\\ &= \mathrm{d}r\sinh^7 r\cosh^7 r\int_r^{\infty} \mathrm{d}(\cosh s)K_m(s,0,y)q_{t,9}(\cosh s).\end{aligned}$$

By changing the variable $\cosh s = \cosh r\cosh u$ for $u \geq 0$, the last expression becomes

$$\mathrm{d}r\sinh^7 r\cosh^7 r\int_0^{\infty} {}_2F_1\left(m+3,-m-3,\frac{1}{2};\frac{1-\cosh u}{2}\right)q_{t,9}(\cosh r\cosh u)\,\mathrm{d}u.$$

Therefore $p_t(r,\eta)$ has the integral representation

$$2\sum_{m\geq 0}\mathrm{e}^{-(m(m+6)+33)t}h_m(\eta)\int_0^{\infty} {}_2F_1\left(m+3,-m-3,\frac{1}{2};\frac{1-\cosh u}{2}\right)q_{t,9}(\cosh r\cosh u)\,\mathrm{d}u.$$

Now, notice that ${}_2F_1\big(m+3,-m-3,\frac{1}{2};\frac{1-\cosh u}{2}\big)$ is simply the Cheybyshev polynomial of the first kind

$$T_{m+3}(x) = {}_2F_1\left(m+3,-m-3,\frac{1}{2};\frac{1-x}{2}\right),$$

for all $x\in\mathbb{C}$. Therefore, one has

$${}_2F_1\left(m+3,-m-3,\frac{1}{2};\frac{1-\cosh u}{2}\right) = T_{m+3}(\cosh u) = \cosh((m+3)u),$$

and the proof is over. □

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